

Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

July 10, 2026

1 Roadmap and rigour status

This lab-book is the *paper-track* of the project: it reproduces the af-validated (T0) results of the registry in `argument/` together with a thin spine of open conjectures that orients them. It is deliberately *not* a mirror of the whole registry — the non-rigorous results (inherited `proved-mod-audit`, `conjecture`, `numerical`, `heuristic`) are anchored once, honestly, in the status ledger of §21 and are never restated as theorems here. The north star is `op-classical`; its status is *open*, and its status-ledger anchor is 1. The definitions used below are the canonical shards `def-stochastic`, `def-almost-idempotent`, `def-near-positive-projection`, `def-signed-idempotent`, `def-negative-mass`, `def-height`, `def-visible-set`, `def-invisible-mass`, and `def-exposed`; this report references those identifiers and introduces no parallel definitions.

The target itself is the classical (stochastic) stability statement: there are universal constants $\eta_0, C > 0$ such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta \leq \eta_0$ admits a stochastic idempotent E with

$$\|Q - E\|_{\infty \rightarrow \infty} \leq C\sqrt{\eta}, \quad (1)$$

with the exponent $\frac{1}{2}$ sharp. This remains **open** in this repo.

The rigorous line is narrow by design. A result may be called rigorous here only when it is byte-matched to a local theorem under `refs/`, validated in an `af` workspace, or Lean-proved. At the time of this shard the registry records **29 af-validated results**. Those reproduced as full sections of this report are the signed-stochastic bridge 1; the halo-collapse family 2, 3, 4, 5, 6, 22, 23, 24, 25; the (EX) /factorization and fan-payment chain 7, 8, 9, 10, 11, 12; the argmin engine 13; the hiddenness witnesses and their consequences 14, 15, 16, 17, 18; the slab/capacity primitives 19, 20, 21; and the starvation obstruction 26. Every other registry result remains anchored only in the status ledger, with its lower-rung status left intact.

The working headline is itself status-sensitive. The inherited campaign distinguishes the realizable-family *linear law* from the worst-case $\sqrt{\cdot}$ -envelope: along the realizable family the tight relation between negative mass δ and height H is

$$\delta = \frac{1}{2} H \quad (\text{linear}),$$

while the quadratic form $\delta \gtrsim H^2$ is only the worst-case envelope, binding because $H \leq O(\tau) = O(\sqrt{\delta})$. The $\sqrt{\eta}$ exponent of (1) is nonetheless sharp, certified by the family `ex-hume` (anchored at 1), which is still `proved-mod-audit` here rather than one of the rigorous results. Crossing between the signed and stochastic pictures should cite the validated bridge 1 rather than re-normalising notation locally.

2 The signed-stochastic bridge

Lemma 1 (Registry contract for lem-classical-equiv). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants: Q row-stochastic with $\|Q^2 - Q\| \leq \eta$ gives $P = \text{theta}(2Q - 1)$ signed affine retraction with $\|P - Q\| \leq C \eta$ and neg mass $\delta \leq C \eta$, and conversely row-normalising p_{i^+} gives Q with $\|P - Q\| \leq 2 \delta$, $\|Q^2 - Q\| \leq 6 \delta + 4 \delta^2$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 29-node adversarial tree with root `validated` and clean taint. The human-readable export is `proofs/lem-classical-equiv/export.md`; the export also records the earlier missing-definition challenge and its repair by importing `def-negative-mass`.

Role in the chain. This bridge is the authorized way to move between `op-classical`'s stochastic formulation and the signed exact-idempotent registry. It is a dependency for the inherited simplex, rank-one, and approximate-simplex reductions, but those downstream reductions remain `proved-mod-audit` unless and until they clear an L0 gate.

3 Height collapse

Proposition 2 (Registry contract for obs-height-collapse). *Height collapse: for an exact signed idempotent P with $0 < \delta(P) \leq 1/4$, nonempty visible set $W(P)$, and hidden top vertex v (height $H = \text{dist}_1(p_v, \text{conv } W)$ maximal among rows), the invisible mass σ_v and the row negative mass ν_v satisfy $H * (1 - \sigma_v) \leq \nu_v * (2 + 4*\delta)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 19-node adversarial tree with root **validated** and taint 19/19 clean. The export is `proofs/obs-height-collapse/export.md`. Earlier compound "hence" clauses were excluded from the contract before validation; this section records only the validated inequality.

Role in the chain. This result explains why invisible mass alone does not finish the Kernel/(EX) route: height is controlled only after pricing the complementary mass. A future dimension-free cap on the relevant invisible mass would combine with this inequality, but that cap is not proved here.

4 Mass split bookkeeping

Lemma 3 (Registry contract for `lem-mass-split`). *Mass split: for an exact signed idempotent P and any row index v , writing $a_j = P_{\{vj\}}$, $a_j^{+} = \max(a_j, 0)$, $a_j^{-} = \max(-a_j, 0)$, and $nu_v = \sum_j a_j^{-}$, one has $\sum_j a_j^{+} = 1 + nu_v$.*

Status. `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a clean 9-node adversarial tree with root `validated` and taint 9/9 `clean`. The export is `proofs/lem-mass-split/export.md`.

Role in the chain. This is the mass bookkeeping dependency factored out of the first `conj-halo-collapse` elevation attempt. Its purpose is to keep later halo-collapse proof trees from re-deriving the same row-sum identity in multiple sibling branches.

5 Convex outsourcing

Lemma 4 (Registry contract for lem-residual-lower). *Convex outsourcing: let C be the convex hull of finitely many points of $\mathbb{R}^n$, and suppose $p = \sum_{j=1}^m c_j p_j + (1 - s) q$ with points p_j, q in $\mathbb{R}^n$, coefficients $c_j \geq 0$, $s = \sum_{j=1}^m c_j < 1$, and $\text{dist}_1(p_j, C) \leq \text{dist}_1(p, C)$ for every j ; then $\text{dist}_1(p, C) \leq \text{dist}_1(q, C)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records 32 validated live nodes, 36 total nodes including 4 archived nodes, root validated, and taint 36/36 clean. The export is proofs/lem-residual-lower/export.md.

Role in the chain. This is a frame-free convex geometry dependency for conj-halo-collapse. It supplies the lower residual-distance comparison used when a hidden row is split into ordinary mass plus a residual term.

6 Residual distance bound

Lemma 5 (Registry contract for lem-residual-upper). *Residual distance bound: let C be the convex hull of finitely many points of R^n , let $b_1..b_M, c_1..c_N \geq 0$ with $m = \sum_j b_j - \sum_k c_k > 0$, let p_j, r_k be points of R^n with $q = (\sum_j b_j p_j - \sum_k c_k r_k) / m$, and let $D_k \geq 0$ satisfy $\|x - r_k\|_1 \leq D_k$ for all x in C and each k ; then $m * \text{dist}_1(q, C) \leq \sum_j b_j * \text{dist}_1(p_j, C) + \sum_k c_k * D_k$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records 49 validated live nodes, 52 total nodes including 3 archived nodes, root validated, and taint 52/52 clean. The export is proofs/lem-residual-upper/export.md. The tree size is above the 12-node brittleness target, so the tracked refactor warning remains honest rather than being hidden in prose.

Role in the chain. This is the companion frame-free convex geometry dependency for conj-halo-collapse. Together with 3 and 4, it packages the residual-recombination estimate needed by the halo-collapse bridge, now itself validated in 6.

7 Halo-robust height collapse

Lemma 6 (Registry contract for conj-halo-collapse). *Halo-robust height collapse:*
*for an exact signed idempotent P with $0 < \delta(P) \leq 1/4$, nonempty visible set $W(P)$, and hidden top vertex v of height H , let σ be the invisible mass of v , σ_g the halo-robust invisible mass (the positive coefficient mass v places on rows at $\ell-1$ distance $> \tau/4$ from $\text{conv } W$, $\tau = \sqrt{\delta(P)}$), and ν_v the row negative mass; then $H * (1 - \sigma_g) \leq (\sigma - \sigma_g) * \tau/4 + \nu_v * (2 + 4*\delta)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 20-node adversarial tree, root validated, taint 20/20 clean (run 2 on the factored workspace). The export is `proofs/conj-halo-collapse/export.md`. The three af-validated dependencies 3, 4, and 5 are imported as externals, so the root rests only on validated or cited inputs.

Elevation history. Run 1 (pre-factoring) ballooned past the node cap (49 live nodes over the cap of 40) and was aborted as a structure signal, not a mathematical refutation: the mass-split book-keeping and the residual-distance estimates were being re-derived inline across siblings. Factoring those into the three registry sub-lemmas above cured it; run 2 closed at 20 nodes.

Role in the chain. This bridge refines the validated collapse bound 2 by pricing recipients inside the $\tau/4$ -halo of the visible hull at their actual distance rather than the worst case H , so the bound stays non-vacuous even when the raw invisible mass exceeds one through self or halo mass (1). Together with a halo-robust cap $\tilde{\sigma}_g \leq 1 - c$ for hidden top vertices it would give $H = O(\tau)$, the height cap of the Kernel Conjecture 1. That cap is OPEN: nothing in this section promotes the kernel chain.

8 Factorization bound

Lemma 7 (Registry contract for lem-factorization). *Factorization bound: let P be an exact signed idempotent (square real matrix with $P^2 = P$ and all row sums equal to 1), let $\delta(P) = \max_i \sum_j \max(-P_{ij}, 0)$, let $k = \text{rank}(P)$, and let $U = (u_1, \dots, u_k)$ be an actual-row basis of P (the rows $p_{\{u_1\}}, \dots, p_{\{u_k\}}$ of P form a basis of the row space of P) whose Gram volume satisfies $\text{Vol}(U) \geq (1/2) * \text{Vol}_{\max}(P)$, where $\text{Vol}_{\max}(P)$ is the maximum Gram volume over all actual-row bases of P ; define coordinates $a_t(j)$ by $p_j = \sum_t a_t(j) p_{\{u_t\}}$, and for each pivot index s in $\{1, \dots, k\}$ set $\beta_s(j) = P_{\{u_s\}j}$, $\lambda_s(j) = 1 - a_s(j)$, $\mu_s(j) = \sum_{t \neq s} \max(-a_t(j), 0)$, $\sigma_s(j) = \sum_{t \neq s} \max(a_t(j), 0)$, $E_s(j) = \max(\mu_s(j) - \lambda_s(j), 0)$, $\Phi_s(U) = \sum_j \max(\beta_s(j), 0) * E_s(j)$, and $S*_s(U) = \sum_j \max(\beta_s(j), 0) * (\sigma_s(j) + 2 * \max(-\lambda_s(j), 0))$; then for every pivot s , $S*_s(U) \leq 2 * \Phi_s(U) + 6 * \delta(P)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records an 11-live-node adversarial tree (12 total nodes, 1 archived), root **validated**, taint 12/12 **clean**, closed in run 1 after five rounds. The export is **proofs/lem-factorization/e**. The contract was narrowed to the single inequality before seeding; the inherited tightness claim for the constants (2,6) was deliberately not elevated and remains **proved-mod-audit** material in the registry shard body.

Role in the chain. This is the composition link of the (EX) program: any universal chart bound $\max_s \Phi_s(U_0) \leq C_0 \delta(P)$ over the $\theta = 1/2$ class now composes rigorously to the class-score constant $C_{sf} = 2C_0 + 6$. The (EX) input itself (1) remains a conjecture: certified numerical evidence supports a plateau near $C_0 = 2$, but nothing in this section promotes it.

9 Zero-sum triangle bound

Lemma 8 (Registry contract for `lem-zerosum-triangle`). *Zero-sum triangle bound:*
let w and v be vectors in $\mathbb{R}^d$ with v having coordinate sum zero ($\sum_l v(l) = 0$);
write $n(x) = \sum_l \max(-x(l), 0)$; then $n(w - v) \leq n(w) + n(v)$.

Status. `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 10-node adversarial tree, root `validated`, taint 10/10 `clean`, closed in run 1 after three rounds. The export is `proofs/lem-zerosum-triangle/export.md`. The run-1 MISSING-fact challenge isolated the load-bearing hypothesis: the identity $\sum_l \max(v(l), 0) = n(v)$ holds exactly because v has zero coordinate sum; for general v only the pointwise inequality survives.

Role in the chain. First factored dependency of 10: it was split out of the `lem-fan-payment` workspace after two balloon aborts forced a linear proof chain into cross-sibling re-derivations. Downstream `af` trees import it as a validated external instead of re-deriving the triangle step per node.

10 Weighted minimum bound

Lemma 9 (Registry contract for `lem-weighted-min`). *Weighted minimum bound: let $p_1, \dots, p_m$ be positive reals with $\sum_i p_i = 1$ and let $n_1, \dots, n_m$ be real numbers; then $\min$ over i in $\{1, \dots, m\}$ of $n_i \leq \sum_i p_i * n_i$.*

Status. `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

Proof tree. The registry shard records an 8-node adversarial tree, root `validated`, taint 8/8 `clean` (the build run's verifier calls were blanked by a network outage and the verify phase was resumed cleanly). The export is `proofs/lem-weighted-min/export.md`.

Role in the chain. Second factored dependency of 10: the index-selection step (some support point has value at most the weighted average). Textbook-adjacent, but registered as its own shard so `af` trees can import it as a validated external instead of re-deriving it across siblings.

11 Zero-sum fan payment

Lemma 10 (Registry contract for lem-fan-payment). *Zero-sum fan payment: let $(w_1, p_1), \dots, (w_m, p_m)$ be a finite family with vectors w_i in $\mathbb{R}^d$ and weights $p_i > 0$ satisfying $\sum_i p_i = 1$, such that every w_i has coordinate sum zero ($\sum_l w_{i,l} = 0$) and the weighted barycenter is zero ($\sum_i p_i w_i = 0$); write $n(w) = \sum_l \max(-w(l), 0)$; then min over i^* in $\{1, \dots, m\}$ of $\sum_i p_i n(w_i - w_{i^*}) \leq 2 * \sum_i p_i n(w_i)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 15-node adversarial tree, root **validated**, taint 15/15 clean, closed in run 3 on the factored workspace; it imports the af-validated dependencies 8 and 9 as externals. Runs 1–2, before the factoring, balloon-aborted at 39 and 47 nodes; the factoring cured the balloon (15 versus 47). The export is **proofs/lem-fan-payment/export.md**.

Role in the chain. The all-mass fan payment lemma is the discrete mechanism behind the certified plateau-2 constant on the reduced fan families of the arm-A campaign: in a reduced edge-fan chart the anchor-pivot score is exactly the weighted average $\sum_i p_i n(w_i - w_*)$, so this lemma bounds the fan-template Φ/δ by 2. The registry shard body additionally records an exact refutation certificate for the over-broad variant (arbitrary w_0 , no zero-sum hypothesis, ratio 5); that certificate is wave material and is not elevated here.

12 Negative-part subadditivity

Lemma 11 (Registry contract for `lem-negpart-subadditive`). *Negative-part subadditivity:*
for all vectors x and y in $\mathbb{R}^d$, writing $n(w) = \sum_l \max(-w(l), 0)$, one has $n(x + y) \leq n(x) + n(y)$.

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 16-node adversarial tree, root `validated`, taint 16/16 clean, closed in run 1. The export is `proofs/lem-negpart-subadditive/export.md`. The lemma was pre-factored *before* elevation, applying the balloon lesson from `lem-fan-payment` instead of re-learning it.

Role in the chain. Pre-factored dependency of 12: the D-restricted fan proof's barycenter step $D \geq n(\sum_{i \in A} p_i w_i)$ uses exactly this pointwise subadditivity (with nonnegative homogeneity, which is definitional).

13 D-restricted zero-sum fan payment

Lemma 12 (Registry contract for lem-fan-payment-restricted). *D-restricted zero-sum fan payment: let $(w_1, p_1), \dots, (w_m, p_m)$ be a finite family with vectors w_i in $\mathbb{R}^d$ and weights $p_i > 0$ satisfying $\sum_i p_i = 1$, such that every w_i has coordinate sum zero and the weighted barycenter is zero ($\sum_i p_i w_i = 0$); write $n(w) = \sum_l \max(-w(l), 0)$; let w_* be a minimizer of $v \rightarrow \sum_i p_i n(w_i - v)$ over $\{w_1, \dots, w_m\}$ and let $A = \{i : n(w_i - w_*) > 0\}$; then $\sum_{i \in A} p_i n(w_i - w_*) \leq (2 + \sqrt{2}) * \sum_{i \in A} p_i n(w_i)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 27-node adversarial tree, root **validated**, taint 27/27 clean, closed in run 1 after ten rounds; it imports the af-validated dependencies 8 and 11 as externals. The export is **proofs/lem-fan-payment-restricted/export.md**. Proof shape: with q the duplicate-pivot mass, horn (i) gives $N \leq D/q$ and horn (ii) gives $N \leq 2(1 - q)/(1 - 2q) D$ for $q < 1/2$; the bounds cross at $q_0 = 1 - 1/\sqrt{2}$ with common value $2 + \sqrt{2}$.

Role in the chain. The D-restricted refinement of 10: the denominator keeps only rows with positive distance to the selected pivot, the fan shadow of the degenerate-set restriction behind **conj-degenerate-payment**. The registry shard body records exact certificates (wave material, not elevated here) refuting constant 2 for this variant and attaining $2 + \sqrt{2}$ in the limit, so the validated constant is sharp. The remaining obstruction toward **conj-degenerate-payment** is the lift from $\theta^{-\frac{1}{2}}$ argmin rows to this fan model.

14 Pivot-removing max-stationarity

Lemma 13 (Registry contract for lem-pivot-removing-move).

Pivot-removing max-stationarity:

let P be a rank-3 exact signed idempotent (square real matrix with $P^2 = P$ and all row sums equal to 1), let $U = (u_0, u_1, u_2)$ be an actual-row chart whose rows $p_{\{u_0\}}$, $p_{\{u_1\}}$, $p_{\{u_2\}}$ form a basis of the row space, let $\text{Vol_max}(P)$ be the maximum Gram volume over actual-row charts, $m_U = \text{Vol}(U)/\text{Vol_max}(P)$, define old coordinates $a_t(i)$ by $p_i = \sum_t a_t(i)p_{\{u_t\}}$, $\beta_r(i) = P_{\{u_r\}i}$, $\lambda_r(i) = 1 - a_r(i)$, $\mu_r(i) = \sum_{t \neq r} \max(-a_t(i), 0)$, $E_r(i) = \max(\mu_r(i) - \lambda_r(i), 0)$, $\Phi_r(U) = \sum_i \max(\beta_r(i), 0)E_r(i)$, and $\Phi(U) = \max_r \Phi_r(U)$, assume U is a theta-half Φ -argmin meaning $m_U \geq 1/2$ and $\Phi(U)$ is minimal among actual-row charts W with $\text{Vol}(W)/\text{Vol_max}(P) \geq 1/2$, fix a maximal pivot s with $\Phi_s(U) = \Phi(U)$, and let $j \notin \{u_0, u_1, u_2\}$ have $c = a_s(j) \neq 0$ such that the pivot-removing chart $V_j = U - u_s + j$ is theta-half admissible, equivalently $|a_s(j)| * m_U \geq 1/2$ with volume factor $\text{Vol}(V_j)/\text{Vol}(U) = |a_s(j)|$; defining new coordinates on V_j by $a_s^j(i) = a_s(i)/c$ and $a_t^j(i) = a_t(i) - a_s(i)a_t(j)/c$ for $t \neq s$, $E_r^j(i) = \max(\sum_{q \neq r} \max(-a_q^j(i), 0) - (1 - a_r^j(i)), 0)$, $\Psi_j = \Phi_s(V_j) = \sum_i \max(P_{\{ji\}}, 0)E_s^j(i)$, and $\Gamma_j = \max_{r \neq s} \Phi_r(V_j) = \max_{r \neq s} \sum_i \max(P_{\{ur\}i}, 0)E_r^j(i)$, one has $\Phi_s(U) \leq \max(\Psi_j, \Gamma_j)$.

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 9-node adversarial tree, root validated, taint 9/9 clean, closed in run 1 after three rounds with zero challenges. The export is `proofs/lem-pivot-removing-move/export.r`. The statement was codified from the arm-G wave record with a fully-inline contract and elevated pre-factored, so the tree stayed small.

Role in the chain. This is the minimality tool of the argmin engine: it is the first statement of the campaign that genuinely uses Φ -minimality of the chart, via the pivot-removing move with exact volume factor $|a_s(j)|$. Every attack on the remaining self-support control (conj-sc and its pivot-removing blocker reduction) consumes this disjunction: an admissible pivot-removing move must be blocked by the new-pivot score Ψ_j or by collateral Γ_j on another pivot. The wave-9 record realizes the volume-inadmissible and Ψ -blocked alternatives by exact certified instances; the collateral alternative is the open branch, and its analysis now rests on this validated statement.

15 The hiddenness dual witness and its always-tight support

Throughout this section P is an exact signed idempotent (`def-signed-idempotent`) with $\delta = \delta(P) > 0$, and we write the derived scales

$$\tau = \sqrt{\delta}, \quad \rho = 4\tau, \quad \kappa = \frac{1}{4}\tau,$$

as fixed in `CONVENTIONS.md` (c). For a row vertex v with row point p_v set $d_i = p_i - p_v$ and let

$$F_v = \{j : \|p_j - p_v\|_1 \geq \rho\}$$

be the ρ -far row-index set. Recall (`def-exposed`) that the *exposedness margin* of v is

$$t^*(v) = \sup_h \min_{f \in F_v} h(p_f), \quad (2)$$

the supremum over affine *admissible exposers* h with $h(p_v) = 0$ and $0 \leq h(p_i) \leq 1$ for every row i ; the vertex v is *hidden* when it is not (ρ, κ) -exposed, i.e. $t^*(v) < \kappa$.

15.1 Existence of the witness

Lemma 14 (`lem-hiddenness-dual-witness`). *Let v be a hidden row vertex of P . Then $F_v \neq \emptyset$ and there exist coefficients $\lambda_f \geq 0$ ($f \in F_v$) with $\sum_f \lambda_f = 1$, and $\alpha_i, \beta_i \geq 0$ over all row indices i with $\sum_i \beta_i = t^*(v) < \kappa$, such that*

$$\sum_{f \in F_v} \lambda_f (p_f - p_v) + \sum_i \alpha_i (p_i - p_v) = \sum_i \beta_i (p_i - p_v). \quad (3)$$

Registry contract (`verbatim`).

Hiddenness dual witness: for an exact signed idempotent P and a hidden row vertex v ($\rho = 4\tau$, $\kappa = \tau/4$, $\tau = \sqrt{\delta(P)}$), writing $F_v = \{j : \|p_j - p_v\|_1 \geq \rho\}$ for the ρ -far row-index set (nonempty for hidden v), there exist $\lambda_f \geq 0$ ($f \in F_v$) with $\sum_f \lambda_f = 1$ and $\alpha_i, \beta_i \geq 0$ (over all row indices i) with $\sum_i \beta_i = t^*(v) < \kappa$, such that $\sum_f \lambda_f (p_f - p_v) + \sum_i \alpha_i (p_i - p_v) = \sum_i \beta_i (p_i - p_v)$.

Proof sketch. Hiddenness first supplies the scalar prerequisites: were F_v empty, `def-exposed` would give $t^*(v) = +\infty$, so a hidden vertex forces $F_v \neq \emptyset$ and $t^*(v) < \kappa$. Writing an admissible exposers as $h(x) = u \cdot (x - p_v)$, the margin (2) is the value of the finite linear program

$$\max_{u, t} t \quad \text{s.t.} \quad 0 \leq u \cdot d_i \leq 1 \quad (\forall i), \quad t \leq u \cdot d_f \quad (\forall f \in F_v), \quad (4)$$

which is feasible at $(u, t) = (0, 0)$ and, since $F_v \neq \emptyset$, bounded above by 1. Dualising (4) with multipliers $\beta_i \geq 0$ on $u \cdot d_i \leq 1$, $\alpha_i \geq 0$ on $-u \cdot d_i \leq 0$, and $\lambda_f \geq 0$ on $t - u \cdot d_f \leq 0$ gives

$$\min \sum_i \beta_i \quad \text{s.t.} \quad \sum_f \lambda_f = 1, \quad \sum_f \lambda_f d_f + \sum_i \alpha_i d_i = \sum_i \beta_i d_i.$$

Finite LP strong duality and dual attainment produce an optimal dual point with $\sum_i \beta_i = t^*(v)$; substituting $d_i = p_i - p_v$ is exactly (3), and $t^*(v) < \kappa$ closes the count. $\square$

Provenance. status: `proved`, `af: validated`; `workspace proofs/lem-hiddenness-dual-witness` (export node 1). First-principles finite LP duality with no imports; a fresh-codex hostile verifier re-derived feasibility, boundedness, attainment, and strong duality from scratch and checked the witness on the banked 7×7 duplicate-split fixture ($t^*(v) = \frac{1}{21} < \frac{1}{16}$).

15.2 The witness support is always tight

Fix a hidden geometrically distinct row vertex u with nonempty visible set $W(P)$. On the primal optimal face of (4) define the *always-tight* constraint families

$$T = \{f \in F_u : h(p_f) = t^*(u) \forall \text{ optima } h\}, \quad O = \{i : h(p_i) = 1 \forall \text{ optima}\}, \quad Z = \{i : h(p_i) = 0 \forall \text{ optima}\}.$$

Lemma 15 (`lem-always-tight-dual-support`). *After deleting redundant centered-zero constraints, every optimal hiddenness dual witness (λ, α, β) of u satisfies*

$$\text{supp}(\lambda) \subseteq T, \quad \text{supp}(\beta) \subseteq O, \quad \text{supp}(\alpha) \subseteq Z,$$

with $T \neq \emptyset$, and $O \neq \emptyset$ if and only if $t^*(u) > 0$.

Registry contract (`verbatim`).

Always-tight dual support: for the exposedness LP at a hidden geometrically distinct row vertex u of an exact signed idempotent P with $\delta(P) > 0$ and nonempty visible set, every optimal hiddenness dual witness (λ, α, β) , after deleting redundant centered-zero constraints, has $\text{supp}(\lambda)$ contained in T , $\text{supp}(\beta)$ contained in O , and $\text{supp}(\alpha)$ contained in Z , where T , O , Z are the rho-far, upper-box, and lower-box constraint families tight on the WHOLE primal optimal face; T is nonempty, and O is nonempty if and only if $t^*(u) > 0$.

Proof sketch. Pair any primal optimum h with the dual balance of Lemma 14. Using $h(p_u) = 0$, $\sum_f \lambda_f = 1$, and $\sum_i \beta_i = t^*(u)$ yields the *complementarity identity*

$$\sum_f \lambda_f (h(p_f) - t^*(u)) + \sum_i \alpha_i h(p_i) + \sum_i \beta_i (1 - h(p_i)) = 0. \quad (5)$$

Every summand of (5) is nonnegative by primal feasibility, so each vanishes: a positive λ_f forces $h(p_f) = t^*(u)$, a positive α_i forces $h(p_i) = 0$, and a positive β_i forces $h(p_i) = 1$. As h ranges over all optima this places the supports in T , Z , and O . Since $\sum_f \lambda_f = 1$ the set $\text{supp}(\lambda)$ is nonempty, so $T \neq \emptyset$; and $O \neq \emptyset$ iff some $\beta_i > 0$ iff $\sum_i \beta_i = t^*(u) > 0$ (when $t^*(u) = 0$ the zero exposer is optimal and no upper box is tight). $\square$

Provenance. status: proved, **af:** validated; workspace proofs/lem-always-tight-dual-support (export node 1), importing Lemma 14 as a validated external. Two independent fresh-codex provers produced the same complementarity core; a separate hostile verifier added the redundant-centered-zero caveat (retained in the “reduced witness” clause) and the $t^* = 0$ edge.

Role in the chain. These two lemmas are the geometric engine of the hiddenness route: Lemma 14 converts “ v is hidden” into a normalised transport identity on the ρ -far rows, and Lemma 15 pins that transport to the tight face. Their consumers are the depth-Markov concentration and far-row existence results of §16. Nothing here bounds the height H ; that is the open cap of the Kernel Conjecture.

16 Consequences of the hiddenness witness

The scales $\tau = \sqrt{\delta}$, $\rho = 4\tau$, $\kappa = \frac{1}{4}\tau$, the far set F_v , and the exposedness margin $t^*(v)$ are those of §15. Write $C = \text{conv}\{p_w : w \in W(P)\}$ and $d_j = \text{dist}_1(p_j, C)$ for the depth of row j ; a hidden top vertex v has height $H = d_v = \max_j d_j$ (**def-height**). The three results below all price the witness of Lemma 14 against an affine 1-Lipschitz functional, using the universal row-separation bound $\|p_i - p_k\|_1 \leq (2 + 4\delta)$ of **def-signed-idempotent**.

16.1 Depth-Markov concentration

Lemma 16 (**lem-hiddenness-depth-markov**). *Let v be a hidden top vertex of height H and (λ, α, β) any hiddenness dual witness of v with $\sum_i \beta_i < \kappa$. Then for every $c > 0$,*

$$\lambda\{f \in F_v : d_f > H - c\tau\} > 1 - \frac{\frac{1}{2} + \delta}{c}. \quad (6)$$

Registry contract (**verbatim**).

Hiddenness depth-Markov: for an exact signed idempotent P with $\text{delta}(P) > 0$, nonempty visible set $W(P)$, hidden top vertex v of height H , and any hiddenness dual witness $(\text{lambda}, \text{alpha}, \text{beta})$ of v with $\text{sum_i } \text{beta_i} < \text{kappa} = \text{tau}/4$ ($\text{tau} = \text{sqrt}(\text{delta})$), one has for every $c > 0$: $\text{lambda}\{f \text{ in } F_v : \text{dist_1}(p_f, \text{conv}\{p_w : w \text{ in } W\}) > H - c*\text{tau}\} > 1 - (\frac{1}{2} + \text{delta})/c$.

Proof sketch. Finite-dimensional ℓ_1 separation at p_v gives an affine functional ϕ with $\phi(p_v) = H$, linear part of ℓ_∞ -norm at most 1, and $\phi(x) \leq \text{dist}_1(x, C)$; set $\psi = H - \phi$, so $\psi(p_v) = 0$. Then for every row i ,

$$0 \leq H - d_i \leq \psi(p_i) \leq \|p_v - p_i\|_1 \leq (2 + 4\delta).$$

Applying the linear part of ψ to the witness balance (3) and using $\psi(p_v) = 0$, $\alpha_i, \beta_i \geq 0$, and $\sum_i \beta_i < \kappa$ gives

$$\sum_f \lambda_f \psi(p_f) \leq \sum_i \beta_i \psi(p_i) < \kappa (2 + 4\delta) = (\tfrac{1}{2} + \delta)\tau,$$

whence $\sum_f \lambda_f (H - d_f) < (\tfrac{1}{2} + \delta)\tau$. Markov's inequality for the nonnegative random variable $H - d_f$ under the probability λ on F_v yields (6). $\square$

Provenance. status: proved, af: validated; workspace **proofs/lem-hiddenness-depth-markov** (export node 1), importing Lemma 14. The hostile verifier *weakened* the hypotheses: the c -parametric bound needs neither tallness nor a δ cutoff, only $\sum_i \beta_i < \kappa$, and it holds for *every* witness tuple.

16.2 A dual certificate for the margin, and a far top-depth row

Lemma 17 (**lem-row-far-dual-certificate**). *Let v be a geometrically distinct row vertex with $\delta > 0$, and set $L_F(v) = \sum_{f: \|p_f - p_v\|_1 \geq \rho} P_{vf}^+$ with $P_{vf}^+ = \max(P_{vf}, 0)$ and ν_v the row- v negative mass. If $L_F(v) > 0$ then*

$$t^*(v) \leq \frac{\nu_v}{L_F(v)}. \quad (7)$$

Registry contract (**verbatim**).

Row-far dual certificate: for an exact signed idempotent P and a geometrically distinct row vertex v with $\delta(P) > 0$ ($\tau = \sqrt{\delta}$, $\rho = 4\tau$), writing $L_F(v) = \sum_{\{f : \|p_f - p_v\|_1 \geq \rho\}} \max(P_{vf}, 0)$ and ν_v the row negative mass, if $L_F(v) > 0$ then $t^*(v) \leq \nu_v / L_F(v)$, where $t^*(v)$ is the exposedness margin of def-exposed.

Proof sketch. For any admissible exposor h , row reproduction $p_v = \sum_j P_{vj} p_j$ with $\sum_j P_{vj} = 1$ and $h(p_v) = 0$ give $0 = \sum_j P_{vj} h(p_j)$. Splitting into sign parts and using $0 \leq h(p_j) \leq 1$ bounds the positive far contribution:

$$\sum_{f \in F_v} P_{vf}^+ h(p_f) \leq \sum_{j: P_{vj} > 0} P_{vj} h(p_j) = \sum_{j: P_{vj} < 0} (-P_{vj}) h(p_j) \leq \nu_v.$$

Dividing the weighted average by $L_F(v) > 0$ gives $\min_{f \in F_v} h(p_f) \leq \nu_v / L_F(v)$ for every h ; taking the supremum over h in (2) yields (7). $\square$

Provenance. status: proved, **af:** validated; workspace proofs/lem-row-far-dual-certificate (export node 1); first-principles weak duality (no attainment needed). Sharp on the banked W29 frontier instance ($\nu_3 / L_F = \frac{1}{81} = t^*(3)$).

Lemma 18 (lem-top-slab-companion). *Let $0 < \delta \leq \frac{1}{2}(17 - 12\sqrt{2})$, let $W(P)$ be nonempty, and let v be a hidden top vertex of height $H > 13\tau$. Then there is a row f with $\|p_f - p_v\|_1 \geq 4\tau$ and*

$$\text{dist}_1(p_f, \text{conv}\{p_w : w \in W\}) > H - \left(\frac{1}{2} + \delta\right)\tau. \quad (8)$$

Registry contract (verbatim).

Top-slab companion: for an exact signed idempotent P with $0 < \delta(P) \leq (17 - 12\sqrt{2})/2$, nonempty visible set $W(P)$, and hidden top vertex v of height $H > 13\tau$ ($\tau = \sqrt{\delta}$), there is a row f with $\|p_f - p_v\|_1 \geq 4\tau$ and $\text{dist}_1(p_f, \text{conv}\{p_w : w \in W\}) > H - (1/2 + \delta)\tau$.

Proof sketch. With the same $\psi = H - \phi$ as above ($0 \leq \psi(p_i) \leq (2 + 4\delta)$), Lemma 14 supplies a witness with $\sum_f \lambda_f \psi(p_f) \leq \sum_i \beta_i \psi(p_i) < \kappa(2 + 4\delta) = (\frac{1}{2} + \delta)\tau$. Since the λ_f are a probability on F_v , some $f \in F_v$ has $\psi(p_f) < (\frac{1}{2} + \delta)\tau$; that f satisfies $\|p_f - p_v\|_1 \geq \rho = 4\tau$ and, via $\phi = H - \psi \leq \text{dist}_1(\cdot, C)$, inequality (8). The constant window $\delta \leq \frac{1}{2}(17 - 12\sqrt{2})$ makes $13 - (\frac{1}{2} + \delta) \geq 4 + 6\sqrt{2} > 4$, so the far row is genuinely deep. $\square$

Provenance. status: proved, **af:** validated; workspace proofs/lem-top-slab-companion (export node 1), importing Lemma 14. The verifier checked the exact symbolic constant chain.

Role in the chain. Lemma 16 shows the witness mass concentrates within an $O(\tau)$ slab below the top, Lemma 17 turns a positive far mass into a quantitative margin bound, and Lemma 18 extracts a single deep far row. Together they are the witness-side inputs to the halo-collapse height cap; the cap itself (1) stays open.

17 Slab and capacity primitives

The three primitives below are elementary consequences of exact idempotence. For an exact signed idempotent P (**def-signed-idempotent**) each row satisfies the *reproduction identity*

$$p_v = \sum_j P_{vj} p_j, \quad \sum_j P_{vj} = 1, \quad (9)$$

so for any affine h with $h(p_v) = 0$ one has $0 = \sum_j P_{vj} h(p_j)$. Write $P_{vj}^+ = \max(P_{vj}, 0)$, $P_{vj}^- = \max(-P_{vj}, 0)$, and $\nu_v = \sum_j P_{vj}^-$ for the row- v negative mass (**def-negative-mass**).

17.1 The Chebyshev low-slab pincer

Lemma 19 (**lem-cs-low-slab-pincer**). *Let v be any row index, let h be affine with $h(p_v) = 0$ and $0 \leq h(p_j) \leq 1$ for every row j , and let $s > 0$. Then*

$$\sum_{j: h(p_j) \geq s} P_{vj}^+ \leq \frac{\nu_v}{s}. \quad (10)$$

Registry contract (verbatim).

CS low-slab pincer: for an exact signed idempotent P , a row index v with $\text{nu}_v = \sum_j \max(-P_{vj}, 0)$, an affine h with $h(p_v) = 0$ and $0 \leq h(p_j) \leq 1$ for every row j , and every $s > 0$, one has $\sum_{j: h(p_j) \geq s} \max(P_{vj}, 0) \leq \text{nu}_v / s$.

Proof sketch. From (9) and $h(p_v) = 0$, the sign split gives $\sum_j P_{vj}^+ h(p_j) = \sum_j P_{vj}^- h(p_j) \leq \sum_j P_{vj}^- = \nu_v$, using $0 \leq h(p_j) \leq 1$. On the slab $A_s = \{j : h(p_j) \geq s\}$, $s P_{vj}^+ \leq P_{vj}^+ h(p_j)$, so $s \sum_{A_s} P_{vj}^+ \leq \sum_j P_{vj}^+ h(p_j) \leq \nu_v$; divide by $s > 0$. $\square$

Provenance. status: proved, **af:** validated; workspace **proofs/lem-cs-low-slab-pincer** (export node 1). The verifier confirmed the minimal hypotheses (no hiddenness, no vertex, no $\delta > 0$) and that equality is attained at $s = t^*$ on the W29 frontier instance, so (10) is sharp.

17.2 Row-zero capacity

Lemma 20 (**lem-row-zero-capacity**). *Let i be a row index and y a vector with $Py = y$, $0 \leq y_j \leq 1$ for all j , $y_i = 0$, and $y_f \geq \kappa$ for all f in a set F . Then*

$$\kappa \sum_{f \in F} P_{if}^+ \leq \nu_i. \quad (11)$$

Registry contract (verbatim).

Row-zero capacity: for an exact signed idempotent P , a row index i , and any vector y with $Py = y$, $0 \leq y_j \leq 1$ for all j , $y_i = 0$, and $y_f \geq \kappa$ for all f in a set F , one has $\kappa \sum_{f \in F} \max(P_{if}, 0) \leq \text{nu}_i$, where nu_i is the row- i negative mass.

Proof sketch. The i -th coordinate of $Py = y$ with $y_i = 0$ gives $0 = \sum_j P_{ij} y_j$, hence $\sum_j P_{ij}^+ y_j = \sum_j P_{ij}^- y_j \leq \sum_j P_{ij}^- = \nu_i$ by $0 \leq y_j \leq 1$. On F , $y_f \geq \kappa$ gives $\kappa \sum_{f \in F} P_{if}^+ \leq \sum_{f \in F} P_{if}^+ y_f \leq \sum_j P_{ij}^+ y_j \leq \nu_i$. $\square$

Provenance. status: proved, **af:** validated; workspace **proofs/lem-row-zero-capacity** (export node 1), resting on Lemma 21 in the registry (which certifies the $Py = y$ hypothesis is the harmonic condition). The verifier made the $y_i = 0$ hypothesis explicit.

17.3 The harmonic–affine bridge

Lemma 21 (`lem-harmonic-affine-bridge`). *For an exact signed idempotent P with rows $p_i = (P_{ij})_j$, a vector g satisfies $Pg = g$ if and only if there is a vector u with $g_i = u \cdot p_i$ for every row index i .*

Registry contract (`verbatim`).

Harmonic-affine bridge: for an exact signed idempotent P with rows $p_i = (P_{ij})_j$, a vector g satisfies $Pg = g$ if and only if there exists u with $g_i = u \cdot p_i$ for every row index i ; in the forward direction $u = g$ works ($g_i = p_i \cdot g$), and the constant term of any affine representation is absorbable into u since all row sums equal 1.

Proof sketch. ($\Rightarrow$) If $Pg = g$ then $g_i = (Pg)_i = p_i \cdot g$, so $u = g$ works. ($\Leftarrow$) If $g_i = u \cdot p_i$ then by row reproduction (9), $(Pg)_i = \sum_j P_{ij}(u \cdot p_j) = u \cdot (\sum_j P_{ij}p_j) = u \cdot p_i = g_i$. A constant term c in an affine representation $g_i = c + u \cdot p_i$ is absorbed by $u' = u + c\mathbf{1}$, since every row sum is 1. $\square$

Provenance. `status:` proved, `af:` validated; workspace `proofs/lem-harmonic-affine-bridge` (export node 1); two lines each direction, checked entrywise on the banked W19 rank-5 fixture.

Role in the chain. Lemma 21 identifies the P -harmonic vectors ($Pg = g$) with affine test functions of the rows, so an admissible expositor is exactly a box-constrained harmonic vector; Lemmas 19 and 20 are then the two capacity estimates that price positive far mass against ν_v . They feed the same halo-collapse and starvation arguments as the witness lemmas of §16.

18 Parametric and depth- d halo collapse

Both results generalise the af-validated halo-collapse bridge 6 by making the halo width a free parameter. Fix an exact signed idempotent P with $0 < \delta \leq \frac{1}{4}$ and nonempty visible set $W(P)$; write $C = \text{conv}\{p_w : w \in W\}$, $d_j = \text{dist}_1(p_j, C)$, $\tau = \sqrt{\delta}$, and $\alpha_j = P_{vj}$ with parts α_j^+, α_j^- and $\nu_v = \sum_j \alpha_j^-$. For a halo width $a > 0$ the mass placed strictly outside the $a\tau$ -halo of C is $\sigma_a = \sum_{j: d_j > a\tau} \alpha_j^+$, and $\sigma = \sum_{j: d_j > 0} \alpha_j^+$ is the invisible mass (`def-invisible-mass`).

18.1 The parametric bridge

Lemma 22 (`lem-parametric-halo-collapse`). *Let v be a hidden top vertex of height H and $a > 0$. Then*

$$H(1 - \sigma_a) \leq (\sigma - \sigma_a)a\tau + \nu_v(2 + 4\delta). \quad (12)$$

Registry contract (`verbatim`).

Parametric halo collapse: for an exact signed idempotent P with $0 < \text{delta}(P) \leq 1/4$, nonempty visible set $W(P)$, hidden top vertex v of height H , and any halo width $a > 0$, writing sigma_a for the positive coefficient mass v places on rows at ell-1 distance $> a*\tau$ from $\text{conv } W$ ($\tau = \text{sqrt}(\text{delta})$), sigma for the invisible mass, and nu_v for the row negative mass, one has $H*(1 - \text{sigma_a}) \leq (\text{sigma} - \text{sigma_a})*a*\tau + \text{nu_v}*(2 + 4*\text{delta})$.

Proof sketch. If $\sigma_a \geq 1$ the left side is ≤ 0 and the right side is ≥ 0 . Otherwise put $A = \{j : d_j > a\tau\}$ and

$$q = \frac{1}{1 - \sigma_a} \left(\sum_{j \notin A} \alpha_j^+ p_j - \sum_k \alpha_k^- p_k \right),$$

so that by row reproduction and mass split ($\sum_j \alpha_j^+ = 1 + \nu_v$, 3) $p_v = \sum_{j \in A} \alpha_j^+ p_j + (1 - \sigma_a)q$ is a genuine convex combination. The lower residual estimate 4 gives $H \leq \text{dist}_1(q, C)$, while the upper residual estimate 5 with $D_k = (2 + 4\delta)$ gives $(1 - \sigma_a) \text{dist}_1(q, C) \leq \sum_{j \notin A} \alpha_j^+ d_j + \nu_v(2 + 4\delta)$. Pricing the non-halo rows by $\sum_{j \notin A} \alpha_j^+ d_j \leq (\sigma - \sigma_a)a\tau$ and multiplying the first bound by $1 - \sigma_a > 0$ yields (12). $\square$

Provenance. `status:` proved, `af:` validated; workspace `proofs/lem-parametric-halo-collapse` (export node 1), importing 3, 4, 5. At $a = \frac{1}{4}$ it recovers the 6 contract `verbatim`; the verifier checked that calibration and the rank-5 fixture.

18.2 The depth- d version

Lemma 23 (`lem-depth-d-halo-collapse`). *Let v be any row index with depth $d = \text{dist}_1(p_v, C)$ and let $a > 0$. With $\sigma_a(v) = \sum_{d_j > a\tau} P_{vj}^+$, $\sigma(v) = \sum_{d_j > 0} P_{vj}^+$, and the deeper-row correction $C_a(v) = \sum_{d_j > a\tau, d_j > d} P_{vj}^+(d_j - d)$,*

$$d(1 - \sigma_a(v)) \leq (\sigma(v) - \sigma_a(v))a\tau + \nu_v(2 + 4\delta) + C_a(v). \quad (13)$$

Registry contract (`verbatim`).

Depth-d halo collapse: for an exact signed idempotent P with $0 < \text{delta}(P) \leq 1/4$ and nonempty visible set $W(P)$, any row index v with $d = \text{dist}_1(p_v, \text{conv}\{p_w : w \text{ in } W\})$, and any halo width $a > 0$, writing $d_j = \text{dist}_1(p_j, \text{conv } W)$, $\text{sigma_a}(v) = \text{sum over } \{j : d_j > a*\tau\} \text{ of } \max(P_{vj}, 0)$ ($\tau = \text{sqrt}(\text{delta})$), $\text{sigma}(v) = \text{sum over}$

$\{j : d_j > 0\}$ of $\max(P_{vj}, 0)$, nu_v the row negative mass, and $C_a(v) = \sum$ over $\{j : d_j > a\tau, d_j > d\}$ of $\max(P_{vj}, 0) \cdot (d_j - d)$, one has $d \cdot (1 - \sigma_a(v)) \leq (\sigma_a(v) - \sigma_a(v)) \cdot a\tau + \text{nu}_v \cdot (2 + 4\delta) + C_a(v)$.

Proof sketch. The trivial branch $\sigma_a(v) \geq 1$ is as before. Otherwise the same residual q (now with $B = \{j : d_j \leq a\tau\}$) gives $p_v = \sum_{j \in A} P_{vj}^+ p_j + (1 - \sigma_a)q$. Convexity of $x \mapsto \text{dist}_1(x, C)$ gives $d \leq \sum_{j \in A} P_{vj}^+ d_j + (1 - \sigma_a) \text{dist}_1(q, C)$; splitting A at $d_j \leq d$ versus $d_j > d$ replaces $\sum_{j \in A} P_{vj}^+ d_j$ by $\sigma_a(v) d + C_a(v)$, so $(1 - \sigma_a)d \leq C_a(v) + (1 - \sigma_a) \text{dist}_1(q, C)$. Pricing $\text{dist}_1(q, C)$ through 5 exactly as in the parametric case gives (13). At a hidden top ($d = H, C_a = 0$) it recovers (12). $\square$

Provenance. status: proved, **af:** validated; workspace `proofs/lem-depth-d-halo-collapse` (export node 1), importing 3, 5. The verifier legitimised the convex split and checked the top-row calibration against 22.

Role in the chain. These are the halo-collapse bounds at arbitrary width and depth: the free parameter a is what lets the genuine-mass bootstrap of §19 choose a halo that separates hidden from visible rows. They tighten but do not close the height cap of the Kernel Conjecture 1.

19 Genuine-mass disintegration and top concentration

Keep the notation of §18: $C = \text{conv}\{p_w : w \in W(P)\}$, $d_j = \text{dist}_1(p_j, C)$, $\tau = \sqrt{\delta}$, height H , and $P_{ij}^+ = \max(P_{ij}, 0)$. Fix a halo width $a > 0$ and the genuine (deep-halo) index set $G_a = \{j : d_j > a\tau\}$, assumed nonempty. The two results below are the “bootstrap” step: they show that the mass a row sends into G_a is either supported on genuinely hidden vertices or is small.

19.1 Disintegration of the halo mass

Lemma 24 (lem-genuine-disintegration). *Fix for every row j a convex vertex representation $p_j = \sum_v \lambda_{jv} p_v$ over geometrically distinct row vertices, and let $g = P\mathbf{1}_{G_a}$. Then every row index i satisfies*

$$g_i \leq M_i^a + \sum_{j \in G_a} P_{ij}^+ \frac{H - d_j}{H - a\tau}, \quad M_i^a = \sum_{j \in G_a} P_{ij}^+ \sum_{v: d_v > a\tau} \lambda_{jv}, \quad (14)$$

where $M_i^a \geq 0$ is positive mass supported entirely on hidden row vertices of depth in $(a\tau, H]$.

Registry contract (verbatim).

Genuine-mass disintegration: for an exact signed idempotent P with $0 < \text{delta}(P) \leq 1/4$, nonempty visible set $W(P)$, height H , halo width $a > 0$ with $G_a = \{j : \text{dist}_1(p_j, \text{conv } W) > a\tau\}$ nonempty ($\tau = \sqrt{\text{delta}}$), fix for every row a vertex representation $p_j = \sum_v \lambda_{jv} p_v$ over geometrically distinct row vertices; then every row index i satisfies $g_i \leq M_i^a + \sum_{j \in G_a} P_{ij}^+ * (H - d_j) / (H - a\tau)$, where $g = P\mathbf{1}_{G_a}$, $d_j = \text{dist}_1(p_j, \text{conv } W)$, $P_{ij}^+ = \max(P_{ij}, 0)$, and $M_i^a = \sum_{j \in G_a} P_{ij}^+ * \sum_{v: d_v > a\tau} \lambda_{jv}$ is positive mass supported entirely on HIDDEN row vertices at depth in $(a\tau, H]$.

Proof sketch. Nonemptiness of G_a forces $H > a\tau$. Applying the upper residual estimate 5 to the vertex representation gives the depth-convexity bound $d_j \leq \sum_v \lambda_{jv} d_v$. Splitting each $j \in G_a$ into shallow mass $L_j = \sum_{d_v \leq a\tau} \lambda_{jv}$ and deep mass $U_j = 1 - L_j$ and using $d_v \leq a\tau$ on the first class, $d_v \leq H$ on the second, yields $d_j \leq a\tau L_j + H U_j$, hence $L_j \leq (H - d_j) / (H - a\tau)$. Since $g_i = \sum_{j \in G_a} P_{ij} \leq \sum_{j \in G_a} P_{ij}^+$ and $1 = L_j + U_j$, weighting the L_j -bound by $P_{ij}^+ \geq 0$ gives (14); the U_j -part is M_i^a , whose vertices have $d_v > a\tau$ and hence are hidden of depth $(a\tau, H]$. $\square$

Provenance. status: proved, af: validated; workspace proofs/lem-genuine-disintegration (export node 1), importing 5. The verifier checked the classification, extreme-point identification, and denominators on the banked rank-3 and rank-5 instances (step 3 of sketch M1).

19.2 Top concentration

Lemma 25 (lem-top-concentration). *Let v be a hidden top vertex of height H and $a > 0$ with $H > a\tau$. The positive mass v places outside the genuine set G_a satisfies*

$$\sum_{j \notin G_a} P_{vj}^+ \leq \frac{\nu_v (2 + 4\delta)}{H - a\tau}. \quad (15)$$

Registry contract (verbatim).

Top concentration: for an exact signed idempotent P with $0 < \text{delta}(P) \leq 1/4$, nonempty visible set $W(P)$, hidden top vertex v of height H , and halo width $a > 0$ with $H > a\tau$ ($\tau = \sqrt{\text{delta}}$), the positive mass v places outside the genuine set satisfies $\sum_{j \notin G_a} P_{vj}^+ \leq \nu_v * (2 + 4\delta) / (H - a\tau)$, where $G_a = \{j : \text{dist}_1(p_j, \text{conv}\{p_w : w \in W\}) > a\tau\}$ and ν_v is the row- v negative mass.

Proof sketch. Choose an ℓ_1/ℓ_∞ support functional $\phi(x) = u \cdot x + b$ with $\|u\|_\infty \leq 1$, $\phi \leq 0$ on C , and $\phi(p_v) = H$; set $t_j = H - \phi(p_j)$. Then $0 \leq t_j \leq (2 + 4\delta)$ (row separation), $t_j \geq H - d_j \geq H - a\tau$ for $j \notin G_a$, and by row reproduction $\sum_j P_{vj} t_j = 0$. The sign split gives $\sum_j P_{vj}^+ t_j = \sum_j P_{vj}^- t_j \leq (2 + 4\delta) \sum_j P_{vj}^- = \nu_v(2 + 4\delta)$. Restricting to $j \notin G_a$, $(H - a\tau) \sum_{j \notin G_a} P_{vj}^+ \leq \sum_j P_{vj}^+ t_j \leq \nu_v(2 + 4\delta)$; divide by $H - a\tau > 0$. $\square$

Provenance. status: proved, af: validated; workspace proofs/lem-top-concentration (export node 1); first-principles support functional, no imports. The verifier ran an exact LP support-functional test on the banked rank-5 instance (step 4 decider).

Role in the chain. Lemma 25 shows a hidden top vertex sends all but an $O(\nu_v/H)$ fraction of its positive mass into the genuine set G_a , and Lemma 24 then re-expresses that genuine mass as mass on strictly hidden vertices plus a controlled shallow remainder. Iterating these against the depth- d collapse 23 is the g -bootstrap intended to force $H = O(\tau)$; the closing cap remains the open Kernel Conjecture 1.

20 The bounded-slab starvation completion obstruction

This standalone obstruction rules out a rank-three completion of the exchange-starvation configuration without any auxiliary constant K . Fix a finite index set I , a real $A \in [4, 6]$, a real $\tau \in (0, \frac{1}{256}]$, and set

$$t = \tau^2, \quad a = \frac{\tau}{1 + \tau}.$$

Lemma 26 (`lem-starvation-completion-obstruction`). *There is no rank-three exact signed idempotent P (so $P^2 = P$, $P\mathbf{1} = \mathbf{1}$, $\text{rank } P = 3$) with row negative mass $\nu_i \leq t$ for every $i \in I$ that carries five distinct full row-point fibers, represented by v, w, f, z, o , such that, writing $D = p_z - p_v$ and $E = p_o - p_v$ (linearly independent),*

$$\|D\|_1 = \tau, \quad p_f - p_v = -AD + tE, \quad p_w - p_v = a(p_f - p_v),$$

with top-row fiber masses $c_v = 1 - \tau$, $c_w = \tau + t$, $c_f = -t$, and $c_Q = 0$ for every other fiber Q , where every full row-point fiber has unique coordinates $p_Q = p_v + x_Q D + y_Q E$ and every nonactor support fiber satisfies $p_Q \in \text{conv}\{p_v, p_w, p_f, p_z, p_o\}$ or $0 \leq y_Q \leq 1$.

Registry contract (`verbatim`).

Bounded-slab starvation completion obstruction (K-free): for every finite I , every real A in $[4, 6]$, every real τ in $(0, 1/256]$, and $t := \tau^2$, $a := \tau/(1+\tau)$, there is no rank-three exact signed idempotent P ($P^2 = P$, $P\mathbf{1} = \mathbf{1}$, $\text{rank } P = 3$) with row negative mass $\nu_i \leq t$ for every i in I having five distinct full row-point fibers represented by v, w, f, z, o such that, with $D := p_z - p_v$ and $E := p_o - p_v$ linearly independent, $\|D\|_1 = \tau$, $p_f - p_v = -A*D + t*E$, $p_w - p_v = a*(p_f - p_v)$, top-row fiber masses $c_v = 1 - \tau$, $c_w = \tau + t$, $c_f = -t$, and $c_Q = 0$ for every other full row-point fiber Q , every full row-point fiber Q has unique reals x_Q, y_Q with $p_Q = p_v + x_Q*D + y_Q*E$, and every nonactor support fiber Q satisfies either p_Q in $\text{conv}\{p_v, p_w, p_f, p_z, p_o\}$ or $0 \leq y_Q \leq 1$.

Proof sketch (by contradiction). Suppose such a P exists. Group the coordinate indices into fibers Q (classes of equal rows) and set $c_Q = \sum_{j \in Q} P_{vj}$, $d_Q = \sum_{j \in Q} (P_{zj} - P_{vj})$, $e_Q = \sum_{j \in Q} (P_{oj} - P_{vj})$. The negative-mass hypothesis $\nu_i \leq t$ gives the subset ledger $-t \leq P_i(S) \leq 1 + t$ and $\|p_i\|_1 \leq 1 + 2t$ for every row i and subset S . Subtracting the v -row from the z -row of $P^2 = P$ gives $DP = D$ with $\sum_j D_j = 0$; expanding $D = \sum_j D_j p_j$ in the coordinates $p_j = p_v + x_j D + y_j E$ and using linear independence of D, E produces the *exact unit moment*

$$\sum_Q x_Q d_Q = 1. \tag{16}$$

Every actor and interior fiber has bounded x -coordinate, $|x_Q| \leq A$, so the hull part contributes at most $A\tau$ (since $\sum_{Q \in \text{Hin}} |d_Q| \leq \|D\|_1 = \tau$). For an exterior fiber Q the disjunction forces $0 \leq y_Q \leq 1$, and the coordinate identity with the row-norm bounds gives the lever estimate $|x_Q|\tau \leq 2 + 4t$; a single global budget from the row- z , row- f , and row- o subset bounds gives $\sum_{Q \in \text{Ext}} |d_Q| \leq t(1 + (2 + t)/A)$, with *no* dependence on the number of fibers. Combining,

$$1 = \sum_Q x_Q d_Q \leq \tau \left[A + (2 + 4t) \left(1 + \frac{2+t}{A} \right) \right].$$

But $0 < \tau \leq \frac{1}{256}$ gives $t < \frac{1}{4}$, $2 + 4t < 3$, $(2 + t)/A \leq 3/4$, and $A \leq 6$, so the right side is strictly below $\frac{1}{256} \left[6 + 3 \left(1 + \frac{3}{4} \right) \right] = \frac{45}{1024} < 1$ — contradicting (16). $\square$

Provenance. status: proved, af: validated; workspace proofs/lem-starvation-completion-obstruction (export node 1). W59 wave: a codex prover paper proof from first principles (guided by, but independent of, the W57/W58 exact Farkas certificates); a fresh hostile codex verifier returned VALID-WITH-CORRECTIONS (a single index-level notation correction applied). Reviewer $\neq$ author (owner: B).

Role in the chain. This is the “K-free” closure of the exchange-starvation leaf: the earlier reductions of that branch depended on an auxiliary completion constant, and this obstruction removes it by showing the rank-three completion cannot exist in the bounded-slab regime. It is the verified four-leaf-system input recorded in the live proof sketch; it does not by itself settle op-classical, which remains open.

21 Status ledger for non-validated registry results

The rows below are anchors, not upgrades. They give every remaining registry result a report label while preserving the registry status. Of the twenty-nine **af**-validated registry results, twenty-six are reproduced as their own sections (enumerated in the roadmap, §1) and are not duplicated in this ledger. The three validated results not yet given a dedicated section (**lem-collateral-import**, **lem-cross-pivot-cancellation**, **lem-import-reduction**) are anchored below at their true **af**-validated status.

Table 1: Registry status ledger for report anchors outside the thirteen validated sections.

Registry id	Status	Contract gloss
conj-degenerate-payment	conjecture	Pivot-local weighted near-degenerate payment bound; the payment horn of the (EX) argmin charge, exact-tested on the wave zoo but unproved.
conj-degenerate-transport	conjecture	Degenerate transverse transport bound; the remaining payment-horn gap after the wave-12 lift, exact-tested but unproved.
conj-ex	conjecture	(EX) chart-bound working form for exact signed idempotents; still a conjectural input equivalent to conj-kernel in the registry.
conj-kernel	conjecture	Kernel Conjecture, the geometric input intended to close op-exposed-hull and hence op-classical .
conj-no-free-frontier	conjecture	Exposedness-absorption statement from the frontier analysis; retained as conjectural and not elevated.
conj-rh	conjecture	Repaired orphan horn: post-fan orphan demand bounded by the unified class/signed/own-negativity budget plus fan residual; exact floor four on the coefficient.
conj-sc	conjecture	Argmin self-support/cancellation control for non-fan beta-positive rows; reduced to the pivot-removing blocker theorem, collateral branch open.
conj-skinny-shadow-cap	conjecture	Skinny two-shadow cap at the root-delta scale, the corrected Route-B statement superseding the retired dual-localization contract.
ex-hume	proved-mod-audit	Explicit 3-by-3 sharpness family for the exponent in op-classical ; inherited proof is not rigorous here.
lem-canonical-separator	proved-mod-audit	Canonical separator package for a hidden top vertex and its harmonic deficit; inherited/mod-audit only.
lem-dual-localization	obstruction (superseded)	Retired 2026-07-04: the transcribed contract is a distance tautology; superseded by conj-skinny-shadow-cap .
lem-exposed-circuit	proved-mod-audit	Concentration and circuit estimate for exposed vertices; inherited/mod-audit only.
lem-collateral-import	proved (af: validated)	Collateral-import bound (CI) for the pivot-removing move, sharp on the wave-10 witness; af-validated in-repo 2026-07-04 (32-node tree, taint clean).
lem-cross-pivot-cancellation	proved (af: validated)	Cross-pivot B-L cancellation identity behind the import anatomy; af-validated in-repo 2026-07-04 (23-node tree, taint clean); near-definitional.

lem-import-reduction	proved (af: validated)	Universal reduction of the collateral import to the cross-pivot masses; af-validated in-repo (G11 reduction (4)); section shard pending.
lem-leakage	proved-mod-audit	Affine-face leakage estimate for almost-idempotent stochastic rows; inherited/mod-audit only.
lem-wiggle-rigidity	proved-mod-audit	F-WR wiggle-rigidity estimate for common-pattern webs; side conditions remain inherited.
obs-deep-leakage	heuristic	Heuristic obstruction showing why a purely local row-identity argument cannot price deep-side mass.
obs-fwr-gap	heuristic	F-WR forbidden-gap obstruction to using web rigidity as the dimension-free shallow-class counter.
obs-linear-law-finite-delta	numerical	Exact finite-delta certificate exceeding the empirical linear-law constant; L3 evidence, not proof.
obs-orphan-amplifier	proved-mod-audit	Exact two-orphan family killing every class/signed-only orphan budget; repaired ratio tends to four; wave paper-proof, orchestrator-recomputed.
obs-sigma-halo-nonrobust	numerical	Exact certificate that the literal zero-halo sigma cap is false; L3 evidence and obstruction.
op-classical	open	North-star classical projection stability problem; still open in this repo.
op-exposed-hull	open	Global exposed-hull lemma that would imply op-classical through the inherited factorization route; open.
prop-approx-simplex	proved-mod-audit	Approximate-simplex reduction to a stochastic idempotent; inherited/mod-audit only.
thm-classical-factorization	proved-mod-audit	Conditional classical factorization theorem through cluster geometry and op-exposed-hull ; not rigorous here.
thm-cluster	proved-mod-audit	Cluster geometry reduction for signed affine retractions; inherited/mod-audit only.
thm-corner-constants	proved-mod-audit	Corner-scale constants, with numerical components explicitly not promoted.
thm-rank-one	proved-mod-audit	Rank-one signed affine retraction stability, including the Hume family; inherited/mod-audit only.
thm-simplex	proved-mod-audit	Simplex and one-dimensional row-polytope stability reduction; inherited/mod-audit only.
thm-well-exposed	proved-mod-audit	Well-exposed-vertex route into the simplex theorem; inherited/mod-audit only.
